

KIPRE STAFF TRAINING NEEDS ASSESSMENT (TNA) REPORT

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For

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Executive Summary

Background

The primary purpose of this Training Needs Analysis (TNA) is to systematically assess institutional competency gaps and evaluate workforce preparedness. This evidence-based approach is designed to inform training priorities and guide strategic human capital development across the KIPRE. The assessment was conducted using a multi-dimensional questionnaire structured into specific modules to capture a holistic view of the workforce. These sections included:

- Basic Information (Directorate, Age, and Sex of the employee)
- Job Role and Responsibilities
- Self Performances Assessing Questions
- Skills and Copetency Assessment
- Training History
- Identification of Training Needs
- Training Delivery Preferences
- Organizational Support
- Future Skills and Emerging Needs
- Additional Comments

By triangulating data from these domains, the analysis identifies not only current skill deficiencies but also the organizational support structures and delivery formats required to bridge them effectively.

Methodology

The assessment was digitized via ODK Central and deployed from 20th – 28th April of 2026. To ensure we captured a representative voice, we used a multi-channel approach—utilizing official email and professional WhatsApp groups. Participation was voluntary, positioned as a collaborative effort to co-design the organization’s future training roadmap. Data Processing & Analysis Once the collection window closed, raw data was ingested into R (v4.5.2). The analytical pipeline focused on cleaning, recoding qualitative responses for statistical depth, and aggregating the results to provide a granular view of needs across all Directorates and Divisions.

Workforce profile

A total of 80 employees, representing 42% of the 189 staff members currently at KIPRE, participated in this assessment. This respondent base spans 23 distinct divisions across seven primary Directorates, ensuring the training roadmap reflects both the core scientific mandate and the essential administrative backbone of the institute. The distribution reveals a workforce concentrated within key technical hubs, most notably the Animal Sciences Welfare and Ethics and Research and Product Development Directorates. In the research domain, the Kenya Snakebite Research and Intervention Division and the Welfare and Ethics Division emerged as the most represented units. This engagement from frontline technical staff—supported by a demographic where 51.2% of respondents are aged between 25 and 44—provides a robust foundation for identifying gaps in laboratory biosafety and modern research protocols. While the gender distribution is

relatively balanced (45% female, 55% male), distinct generational trends are evident: the female cadre is most concentrated in the 35–44 age bracket (36.1%), whereas the male workforce peaks in the 45–54 bracket (34.1%). Administrative and strategic functions are equally well-mapped, with consistent representation from Corporate Services and the Office of the Director General. Notably, only 10% of respondents are aged 55 or older, indicating a predominantly mid-career workforce with significant long-term growth potential. This demographic profile highlights a strategic opportunity to implement cross-cutting training programmes that standardise administrative excellence and ensure the institution is prepared for future succession and evolving digital demands.

Table 1: Workforce distribution by directorate and division

Division / Section	Staff Count
Animal Sciences Welfare And Ethics Directorate	
Kenya Snakebite Research And Intervention Division	1
Laboratory Services Section	1
National Laboratory Animal Breeding Division	4
Veterinary Clinical And Diagnostics Services Division	9
Welfare And Ethics Division	11
Capacity Building Partnership And Grant Management Directorate	
Human Resource And Administration Division	1
Partnerships And Grants Management Division	3
Corporate Services Directorate	
Corporate Communication Division	4
Finance And Accounts Division	4
Human Resource And Administration Division	5
Information Communication Technology Division	1
Planning And Strategy Division	1
Corporation Secretary Legal Services Directorate	
Infectious Diseases And One Health Division	1
Internal Audit And Risk Assurance Directorate	
Human Resource And Administration Division	1
Office Of The Director General	
Human Resource And Administration Division	1
Planning And Strategy Division	1
Research And Product Development Directorate	
Data Science And Analytics Section	1
Evolutionary Ecology Conservation And Ecosystem Health Division	7
Infectious Diseases And One Health Division	7
Kenya Snakebite Research And Intervention Division	10
Laboratory Services Section	1
Non Communicable Diseases Division	1
Reproductive Health And Biology Division	4

Table 2: Gender distribution across age brackets

Gender	Under 25 years	25-34 years	35-44 years	45-54 years	55+ years	Total
female	8.3% (3)	22.2% (8)	36.1% (13)	27.8% (10)	5.6% (2)	100.0% (36)
male	6.8% (3)	29.5% (13)	15.9% (7)	34.1% (15)	13.6% (6)	100.0% (44)
Total	7.5% (6)	26.2% (21)	25.0% (20)	31.2% (25)	10.0% (8)	100.0% (80)

Table 3: Job grade distribution by length of service

Job_Grade	Less Than 1 Year	1 3 Years	4 7 Years	8 15 Years	16+ Years	Total
NM1	0	1	0	0	0	1
NM2	0	1	0	0	0	1
NM5	0	0	0	1	3	4
NM6	0	0	0	0	2	2
NM7	0	3	2	2	3	10
NM8	4	3	9	3	4	23
NM9	0	0	0	1	1	2
NM10	0	1	2	4	5	12
NM11	0	0	1	1	8	10
NM12	2	5	1	1	2	11
NM13	0	2	0	1	1	4

1. Which critical competency gaps exist across directorates, divisions, and job grades, and which areas require immediate training intervention?

This question helps management identify

- High-priority skill deficiencies
- Departments with the greatest capacity gaps
- Staff categories requiring urgent support

Likely outputs

- Competency gap heatmaps
- Directorate ranking tables
- Priority training matrix
- Skill deficiency frequencies

Likely recommendations

- Targeted capacity-building programmes
- Directorate-specific training plans
- Prioritised budget allocation

```
# Select, clean, and reshape data for the gap analysis
gap_analysis_data <- data %>%
```

```

select(
  directorate = section_a.directorate,
  division = section_a.division_section,
  job_grade = section_a.job_grade,
  tech = section_d.tech_skill,
  ict = section_d.ict_skill,
  comm = section_d.comm_skill,
  leadership = section_d.leadership_management,
  policy = section_d.policy_analysis_and_implementation,
  finance = section_d.finance_management,
  customer = section_d.customer_service,
  m_and_e = section_d.monitoring_evaluation
)

competencies <- c(
  "tech", "ict", "comm", "leadership",
  "policy", "finance", "customer", "m_and_e"
)

```

```
gap_long <- gap_analysis_data %>%
```

```

pivot_longer(
  cols = all_of(competencies),
  names_to = "competency",
  values_to = "level"
) %>%

```

```

mutate(
  level = str_to_lower(level),

  gap_score = case_when(
    level == "high" ~ 0,
    level == "medium" ~ 1,
    level == "low" ~ 2,
    TRUE ~ NA_real_
  )
)

```

```
overall_gap_summary <- gap_long %>%
```

```

group_by(competency, level) %>%

```

```

summarise(
  n = n(),
  .groups = "drop"
) %>%

group_by(competency) %>%

mutate(
  percent = round(100 * n / sum(n), 1)
) %>%

arrange(desc(percent))

```

```

directorate_gap <- gap_long %>%

group_by(directorate, competency) %>%

summarise(
  mean_gap = mean(gap_score, na.rm = TRUE),
  low_pct  = mean(level == "low") * 100,
  med_pct  = mean(level == "medium") * 100,
  .groups = "drop"
) %>%

arrange(desc(mean_gap))

directorate_gap %>%

ggplot(aes(
  x = competency,
  y = directorate,
  fill = mean_gap
)) +

geom_tile(color = "white") +

geom_text(
  aes(label = round(mean_gap, 2)),
  size = 3.5,
  color = "black"
) +

```

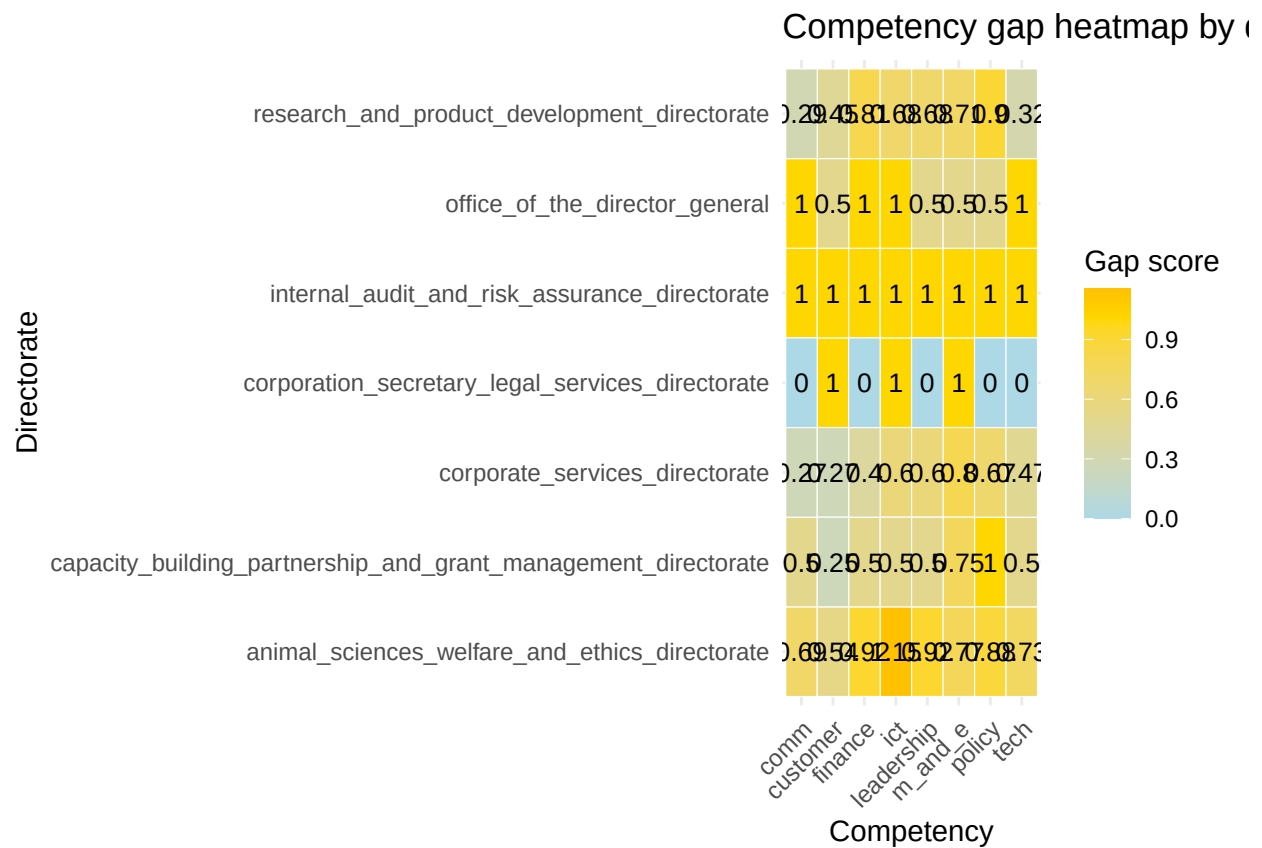
```

labs(
  title = "Competency gap heatmap by directorate",
  x = "Competency",
  y = "Directorate",
  fill = "Gap score"
) +

theme_minimal() +

theme(
  axis.text.x = element_text(
    angle = 45,
    hjust = 1
  )
)+scale_fill_gradient2(
  low = "lightblue",
  mid = "gold",
  high = "red",
  midpoint = 1
)

```



```
division_gap <- gap_long %>%

  group_by(division, competency) %>%

  summarise(
    mean_gap = mean(gap_score, na.rm = TRUE),
    .groups = "drop"
  ) %>%

  arrange(desc(mean_gap))
```

```
job_grade_gap <- gap_long %>%

  group_by(job_grade, competency) %>%

  summarise(
    mean_gap = mean(gap_score, na.rm = TRUE),
    low_pct = mean(level == "low") * 100,
    .groups = "drop"
  ) %>%

  arrange(desc(mean_gap))
```

2. To what extent are employees adequately prepared to meet emerging organizational, technological, and policy-related demands?

- Future workforce readiness
- Digital transformation preparedness
- Emerging competency requirements
- Institutional adaptability

Likely outputs

- Emerging skills rankings
- Workforce readiness summaries
- Future skills dashboard
- Trend analysis

Likely recommendations

- Future-focused training programmes
- Digital skills initiatives
- Leadership and innovation training
- Strategic workforce development planning

Key recommendations

Priority actions

Conclusion

Strategic implications

7. Exemplary visualisations